Naive Bayes Classifier

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Naive Bayes Classifier

Naive Bayes Classifier is one of the simple and most effective Classification algorithms which helps in building the fast machine learning models that can make quick prediction.

1. It is a probabilistic classifier, which means it predicts on the basis of the probability of an object.
2. Some popular examples of **Naive Bayes Algorithm** are **spam filtration**, **Sentimental analysis, and classifying articles**.

Bayes Theorem:

Bayes' theorem is also known as **Bayes' Rule** or **Bayes' law**, which is used to determine the probability of a hypothesis with prior knowledge. It depends on the conditional probability.

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Where,

P(A|B) is Posterior probability: Probability of hypothesis A on the observed event B.

P(B|A) is Likelihood probability: Probability of the evidence given that the probability of a hypothesis is true.

P(A) is Prior Probability: Probability of hypothesis before observing the evidence.

P(B) is Marginal Probability: Probability of Evidence.

Formulas (Bayes Theorem):

$P(A|B) = (P(B|A) * P(A))/ P(B) == P(A \cap B) / P(B)$

$P(B|A) = (P(A|B) * P(B))/ P(A) == P(B \cap A) / P(A)$

if $P(A \cap B) == P(B \cap A)$.

then $P(B|A) * P(A) = P(A|B) * P(B)$

Types Of Naive Bayes:

Gaussian: It is used in classification and it assumes that features follow a normal distribution.

Multinomial: It is used for discrete counts.

Bernoulli: The binomial model is useful if your feature vectors are binary (i.e. zeros and ones).

EXAMPLE-1 [2025_MIDSEM_SPRING]

Consider the following dataset:

Income Level	Credit Score	Loan Amount	Default
Low	Bad	Small	Yes
Low	Bad	Large	Yes
Medium	Average	Medium	No
High	Good	Large	No
High	Average	Medium	No
Medium	Good	Small	No
Low	Average	Medium	Yes
High	Bad	Large	No
Low	Good	Medium	No

Using the Naive Bayes classifier, determine whether a new customer with the following attributes is likely to default on a loan: Income Level = Low, Credit Score = Average, Loan Amount = Medium.

EXAMPLE-1 [SOLUTION]

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Step 1: Prior Probabilities

Total records = 9

Default = Yes → 3 records

Default = No → 6 records

P(Yes) = 3/9 = 1/3

P(No) = 6/9 = 2/3

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Step 2: Likelihoods

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For Default = Yes:

P(Income=Low | Yes) = 3/3 = 1

P(Credit=Average | Yes) = 1/3

P(Loan=Medium | Yes) = 1/3

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For Default = No:

P(Income=Low | No) = 1/6

P(Credit=Average | No) = 2/6 = 1/3

P(Loan=Medium | No) = 4/6 = 2/3

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Step 4: Decision

Since P(Yes | X) > P(No | X),The customer is predicted to DEFAULT.

Final Answer: Default = Yes

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Step 3: Posterior Probabilities

P(Yes | X) = (1/3) × 1 × (1/3) × (1/3) = 1/27

P(No | X) = (2/3) × (1/6) × (1/3) × (2/3) = 2/81

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EXAMPLE-2 [2024_MIDSEM_SPRING]

The following table gives a data set about species. Using Naive Bayes classifier, identify the species of an entity with the following attributes. X={Color=Green, Legs=2, Height=Tall, Smelly=No}

Color	Legs	Height	Smelly	Species
White	3	Short	Yes	M
Green	2	Tall	No	M
Green	3	Short	Yes	M
White	3	Short	Yes	M
Green	2	Short	No	H
White	2	Tall	No	H
White	2	Tall	No	H
White	2	Short	Yes	H

EXAMPLE-3 [2025_MIDSEM_SPRING_SPECIAL-2]

You are given the following dataset

Color	Legs	Height	Smelly	Species
White	3	Short	Yes	M
Green	2	Tall	No	M
Green	3	Short	Yes	M
White	3	Short	Yes	M
Green	2	Short	No	H
White	2	Tall	No	H
White	2	Tall	No	H
White	2	Short	Yes	H

Use naive bayes to check species for test data X = {Color=Green, Legs=2, Height=Tall, Smelly=No}. [5 Marks]

EXAMPLE-2 & 3 [SOLUTION]

Step 1: Prior Probabilities

Total records = 8

Species M = 4

Species H = 4

P(M) = 4/8 = 0.5

P(H) = 4/8 = 0.5

Step 2: Likelihoods

For Species = M:	For Species = H:
P(Color=Green | M) = 2/4 = 0.5	P(Color=Green | H) = 1/4 = 0.25
P(Legs=2 | M) = 1/4 = 0.25	P(Legs=2 | H) = 4/4 = 1
P(Height=Tall | M) = 1/4 = 0.25	P(Height=Tall | H) = 2/4 = 0.5
P(Smelly=No | M) = 1/4 = 0.25	P(Smelly=No | H) = 3/4 = 0.75

Step 4: Decision

Since P(H | X) > P(M | X),

Predicted Species = H

Step 3: Posterior Probabilities

$P(M \mid X) = 0.5 \times 0.5 \times 0.25 \times 0.25 \times 0.25 = 0.0039$

$P(H \mid X) = 0.5 \times 0.25 \times 1 \times 0.5 \times 0.75 = 0.0469$

EXAMPLE-4 [2024_MIDSEM_AUTUMN]

Suppose you are building a Naive Bayes classifier to predict whether an email is **spam** or **not spam** based on the presence of three keywords: "**discount**", "**offer**", and "**sale**". You have collected a training dataset containing 200 emails, out of which 100 are labeled as spam and 100 as non-spam. For the given dataset, you have calculated the following probabilities:

P(spam) = 0.5
P(not spam) = 0.5
P(discount|spam) = 0.8
P(offer|spam) = 0.7
P(sale|spam) = 0.6
P(discount|not spam) = 0.3
P(offer|not spam) = 0.2
P(sale|not spam) = 0.1

Now, given a new email containing the words "discount" and "offer", what is the probability that this email is classified as spam by the Naive Bayes classifier? [3 Marks]

EXAMPLE-5 [2025_ENDSEM_SPRING]

Apply the Naïve Bayes classification algorithm to a given dataset with two features and a binary class label. Compute the prior probabilities, likelihoods, and posterior probabilities to classify a new sample. Justify the classification decision with appropriate mathematical formulations. predict the class label for a new data point:

If $(X_1 = 6, X_2 = 16)$, then $Y = ?$

Sample_ID	Feature_1 (X_1)	Feature_2 (X_2)	Class (Y)
1	5	12	A
2	7	15	B
3	6	14	A
4	8	18	B
5	5	10	A
6	9	20	B
7	6	13	A
8	7	17	B

EXAMPLE-6

Given Data [Outlook=Sunny, Temp=Cool, Humidity=Normal, Wind=Strong], Apply NB Classifier to find the class label of the unknown data.

Day	*Outlook*	*Temperature*	*Humidity*	*Wind*	*PlayTennis*
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

EXAMPLE-7

Consider the given Dataset ,Apply Naive Baye's Algorithm and Predict that if a fruit has the following properties then which type of the fruit it is
Fruit = {Yellow , Sweet ,long}

Frequency Table:

Fruit	Yellow	Sweet	Long	Total
Mango	350	450	0	650
Banana	400	300	350	400
Others	50	100	50	150
Total	800	850	400	1200

Example-7 (Solution)

P(A|B) = (P(B|A) * P(A))/ P(B)

1. Mango:

P(X | Mango) = P(Yellow | Mango) * P(Sweet | Mango) * P(Long | Mango)

a)P(Yellow | Mango) = (P(Mango | Yellow) * P(Yellow))/ P (Mango)

= ((350/800) * (800/1200)) / (650/1200)

P(Yellow | Mango)= 0.53 →1

1.b)P(Sweet | Mango) = (P(Sweet | Mango) * P(Sweet))/ P (Mango)

= ((450/850) * (850/1200)) / (650/1200)

P(Sweet | Mango)= 0.69 → 2

Example-7 (Solution)

c)P(Long | Mango) = (P(Long | Mango) * P(Long))/ P (Mango)

= ((0/650) * (400/1200)) / (800/1200)

P(Long | Mango)= 0 → 3

On multiplying eq 1,2,3 ==> P(X | Mango) = 0.53 * 0.69 * 0

P(X | Mango) = 0

Example-7 (Solution)

2. Banana:

P(X | Banana) = P(Yellow | Banana) * P(Sweet | Banana) * P(Long | Banana)

2.a) P(Yellow | Banana) = (P(Banana | Yellow) * P(Yellow))/ P (Banana)

= ((400/800) * (800/1200)) / (400/1200)

P(Yellow | Banana) = 1 → 4

2.b) P(Sweet | Banana) = (P(Banana | Sweet) * P(Sweet))/ P (Banana)

= ((300/850) * (850/1200)) / (400/1200)

P(Sweet | Banana) = .75→ 5

Example-7 (Solution)

2.c)P(Long | Banana) = (P(Banana | Yellow) * P(Long))/ P (Banana)

= ((350/400) * (400/1200)) / (400/1200)

P(Yellow | Banana) = 0.875 → 6

On multiplying eq 4,5,6 ==> P(X | Banana) = 1 * .75 * 0.875

P(X | Banana) = 0.6562

Example-7 (Solution)

3. Others:

P(X | Others) = P(Yellow | Others) * P(Sweet | Others) * P(Long | Others)

3.a) P(Yellow | Others) = (P(Others| Yellow) * P(Yellow))/ P (Others)

= ((50/800) * (800/1200)) / (150/1200)

P(Yellow | Others) = 0.34→ 7

3.b) P(Sweet | Others) = (P(Others| Sweet) * P(Sweet))/ P (Others)

= ((100/850) * (850/1200)) / (150/1200)

P(Sweet | Others) = 0.67 → 8

Example-7 (Solution)

3.c) P(Long | Others) = (P(Others| Long) * P(Long))/ P (Others)

= ((50/400) * (400/1200)) / (150/1200)

P(Long | Others) = 0.34 → 9

On multiplying eq 7,8,9 ==> P(X | Others) = 0.34 * 0.67* 0.34

P(X | Others) = 0.07742

So finally from P(X | Mango) == 0 , P(X | Banana) == 0.65 and P(X| Others) == 0.07742.

We can conclude Fruit{Yellow,Sweet,Long} is Banana.

EXAMPLE-8

Given the training data in the table (Buy Computer data), predict the class of the following new example using Naïve Bayes classification: age less than or equal to 30, in the given the training data in the table (Buy Computer data), predict the class of the following new example using Naïve Bayes classification: age less than or equal to 30, income=medium, student=yes, credit-rating=fair

age	income	student	credit_rating	buys_computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31…40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31…40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31…40	medium	no	excellent	yes
31…40	high	yes	fair	yes
>40	medium	no	excellent	no

EXAMPLE-9

Consider a football game between two rival teams, say team A and team B. Suppose team A wins 65% of the time and team B wins the remaining matches. Among the games won by team A, only 35% of them comes from playing at team B's football field. On the other hand, 75% of the victories for team B are obtained while playing at home.

1. If team B is to host the next match between the two teams, what is the probability that it will emerge as the winner?
2. If team B is to host the next match between the two teams, who will emerge as the winner?